

A control framework for topological data analysis of perturbation single-cell RNA-seq: dispersion-matched nulls, traversal tests, and a drug-tolerance case study

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¹Preprint — methods / cautionary report. Code:
<https://github.com/htlin222/sctda-cancer-plasticity>

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Abstract

Persistent homology is increasingly applied to single-cell RNA-seq (scRNA-seq) to detect “cyclic” or non-tree-like cell-state structure, with the maximum degree-1 persistence ($\max\text{-}H_1$) used as a scalar readout of cell-state plasticity. We show that this statistic, computed without structure-aware controls, conflates genuine topology with two mundane confounds—transcriptional dispersion and the cell cycle—and we provide a reusable four-part control framework that separates them: (i) a **covariance-matched Gaussian null** that distinguishes real structure from dispersion; (ii) a **circular-coordinate traversal test** that asks whether a detected “loop” is actually occupied before any cyclic interpretation; (iii) **cell-cycle regression evaluated within the dispersion-controlled frame**, which we show is necessary because the common control (raw $\max\text{-}H_1$ preserved after regression) is inadequate; and (iv) **bootstrap directional testing**, because $\max\text{-}H_1$ point estimates are subsample-unstable. On synthetic data with known ground truth the framework is sensitive (it passes a genuine traversed loop) and specific (it rejects benign dispersion and non-Gaussian non-cyclic structure). Applied as a worked case study to five longitudinal EGFR-mutant lung-cancer systems, the framework decomposes an apparently compelling, cross-system, drug-associated $\max\text{-}H_1$ signal: cells never traverse the loop (Rayleigh $R \geq 0.965$ vs 0.881 for a 12%-occupied control), baseline $\max\text{-}H_1$ equals a dispersion-matched null in every system, the in-vitro drug excess is contributed by proliferating cells and vanishes after cell-cycle regression, and the in-vivo (PDX) excess is genuine non-cell-cycle structure that nonetheless still fails the traversal test. In no system does $\max\text{-}H_1$ evidence a cell-traversed cycle.

Contributions.

1. A **covariance-matched Gaussian null** for $\max\text{-}H_1$ separating genuine structure from transcriptional dispersion, superseding the field-standard within-cell gene-shuffle null.
2. A **circular-coordinate traversal test** establishing occupancy of a persistent loop as a precondition for any “cyclic” claim.
3. The observation and correction that the **common cell-cycle control (raw $\max\text{-}H_1$ preserved after S/G2M regression)** is inadequate, because raw $\max\text{-}H_1$ is dispersion-dominated.
4. A **synthetic benchmark** characterising sensitivity and specificity, and a five-system cancer **case study** of a confounded signal that is reproducible, cross-scale, and wrong.

1 Introduction

Topological data analysis (TDA), and persistent homology in particular, is an increasingly popular lens on scRNA-seq. Its appeal is the ability to detect closed, non-tree-like structure (loops; degree-1 homology, H_1) that trajectory-inference tools, designed for one-directional progressions, are not built to quantify. In cancer drug tolerance this is attractive: lineage-tracing studies show drug-tolerant persister cells reversibly transition among transcriptional states, a behaviour naturally described as “cyclic.” A common move is to summarise the topology with a single scalar, $\max\text{-}H_1$, and to read its increase under drug as a measure of cyclic plasticity.

This inherits a known hazard. Persistent homology on a noisy high-dimensional point cloud reports H_1 features whether or not the population occupies or traverses a loop, and $\max\text{-}H_1$ as a scalar conflates genuine structure with two confounds: transcriptional dispersion (the variance/spread of the embedding, which directly inflates Vietoris–Rips persistence) and the cell cycle, which is itself intrinsically a loop ($G1 \rightarrow S \rightarrow G2M \rightarrow G1$). Cautionary methods work has repeatedly shown that single-cell analyses require structure-aware nulls—for doublets, ambient RNA, batch effects, and over-interpreted pseudotime—and topological summaries are no exception. Yet the nulls in common use for scTDA (within-cell gene shuffles) do not control for dispersion, and cell-cycle controls are typically evaluated on raw persistence rather than relative to a dispersion baseline. We provide a control framework that makes both confounds explicit, validate it on synthetic ground truth, and apply it to five EGFR-mutant lung-cancer systems.

2 Results

The control framework

The framework takes a cell-by-PC embedding for each condition and reports three quantities plus a decision rule. (1) *Dispersion test*: the ratio of observed $\max\text{-}H_1$ to that of a multivariate Gaussian matched to the embedding’s mean and covariance; a ratio near 1 means the statistic is explained by dispersion, a ratio robustly > 1 indicates structure beyond dispersion. (2) *Traversal test*: the Rayleigh concentration R of circular coordinates from the most-persistent H_1 class; R near 1 means the population sits at one angle (loop not occupied). (3) *Cell-cycle test*: the dispersion ratio recomputed after S/G2M-score regression. All three are wrapped in bootstrap subsampling. Decision rule: *structure beyond dispersion* if the Gaussian ratio is robustly > 1 ; *a traversed loop* only if additionally $R < 0.6$; *cell-cycle driven* if the excess does not survive S/G2M regression.

The framework is sensitive and specific on synthetic ground truth

We validated the framework on point clouds where the answer is known (Fig. 3). A pure Gaussian blob gives ratio 0.93, $R = 0.99$ (correctly *no structure*). A uniformly traversed loop gives ratio 8.35, $R = 0.42$ (correctly *a traversed loop*; sensitivity). A sparse non-closed arc gives ratio 1.54 but $R = 0.96$ (correctly *structure, not traversed*). Two well-separated Gaussian clusters—non-Gaussian but non-cyclic—give ratio 0.93 (correctly *no structure*; specificity): the Gaussian null does *not* false-positive on benign non-Gaussianity, the key objection to a covariance-matched null. A second non-parametric dispersion-preserving null (independent per-PC permutation) agrees and is more conservative.

Case study: an apparent topological signal of drug tolerance

We computed $\max\text{-}H_1$ (top-30 PCs, $\mathbb{F}_2$) on five EGFR-mutant systems. $\max\text{-}H_1$ increased with drug in the osimertinib cell line and PDX models (osimertinib D0→D14: $1.41 \rightarrow 3.78$; PDX YU-006 untreated→residual: $2.79 \rightarrow 5.46$; Fig. 1), the kind of trend that motivates scTDA

applications. The erlotinib series is a counter-example—weak and non-monotonic (D9 1.77, D11 1.08, the lowest of the series)—which we keep in view as evidence of instability.

Control 1 — cells do not traverse the loop

Circular coordinates collapse to a single angle in every cohort: Rayleigh $R \geq 0.965$ (Fig. 2a)—more concentrated than the $R = 0.881$ of a 12%-occupied positive control. Using Watermelon lineage barcodes, clonal lineages are if anything *more* concentrated than random groups. The persistent loop is a sparse feature the population does not occupy or traverse.

Control 2 — baseline max- H_1 is dispersion

Against the covariance-matched Gaussian null, baseline max- H_1 does not exceed dispersion (osimertinib D0/D3 ratio ≈ 0.9 ; observed $>$ null in 18–31% of subsamples; Fig. 1). An excess appears only under drug (osimertinib D7/D14 in 100%/98%; PDX residual 99%). The field-standard gene-shuffle null, by destroying all covariance, tests only “is there any structure,” not “structure beyond dispersion.” The Gaussian null’s failure mode is conservative for our negative conclusion.

Control 3 — the in-vitro excess is the cell cycle; the in-vivo excess is not

Localising the cells contributing the osimertinib D14 excess (leave-one-cluster-out), the structure-driving population is proliferating cells (DE: PTTG1, UBE2S, CKS1B, MYBL2; proliferation markers MKI67/PCNA/CCNB1/CDK1/BIRC5, 5/6; EMT 1/19). In the pooled embedding, the excess does not survive S/G2M regression: the D14 ratio falls $1.18 \rightarrow 0.99$ and $D14 > D0$ holds in only 31% of bootstraps (Fig. 2b). This exposes a methodological error: the usual control reports *raw* max- H_1 preserved after regression, but raw max- H_1 is dispersion-dominated; the cell-cycle contribution is visible only as a ratio to the dispersion null. The framework decomposes rather than uniformly debunks: in the in-vivo PDX the residual-disease excess (the strongest in the study) is *not* cell cycle—excluding cycling cells leaves it intact (G1-only ratio 2.07, 90% CI [1.37, 2.61], > 1 in 100% of bootstraps). Yet this structure still fails the traversal test ($R = 0.965$): even where structure is real, the population does not traverse it, so the “cyclic plasticity” reading remains unsupported. Across all five systems: dispersion accounts for the baseline; the cell cycle for the in-vitro excess; the in-vivo residual is genuine but non-traversed—and in no system does max- H_1 evidence a cell-traversed cycle.

A confound-resistant clonal statistic is under-powered

We defined clonal state-memory $M(t) = 1 - H_{\text{obs}}/H_{\text{null}}$ against a lineage-shuffle null. A baseline reduction is *suggested* but not robust ($M(D0)$ point estimate 0.48, 90% CI [−0.05, 0.29]), and its decay under drug is not robust: the sign flips across clustering settings (positive in 3/9), the trajectory is non-monotonic, and a depth-stratified null halves it (Fig. 4). The cause is power: only 12–33 clones with ≥ 3 cells per timepoint. Clonal-memory statistics require clone-rich designs current drug-timecourse datasets rarely provide.

Beyond rejection: the framework isolates a genuine, interpretable program

A control framework earns trust only if it passes real structure as well as rejecting confounded structure (cf. the synthetic true positive). The PDX residual is the worked example: its excess is the one signal surviving every control, and the structure-driving cells (leave-one-cluster-out) are a coherent multiciliated airway-epithelial program (PIFO, RSPH1, CFAP45/126/157, CAPS/CAPSL, CETN2, TPPP3; secretory AGR2/AGR3). It is drug-emergent in YU-006—0.1% of untreated cells but 9.6% under residual disease ($\approx 6 \rightarrow \approx 187$ cells in absolute terms while the tumour contracts threefold)—consistent with the differentiated lineage states EGFR-TKI persisters are known to

adopt. The framework thus isolates genuine structure and localises it to an interpretable state, not merely subtracts artefacts. Two bounds: the program is model-specific (absent in YU-003), and copy-number inference could not resolve tumour-derived (transdifferentiation) vs an expanded non-malignant population without a matched-normal reference. And this real structure still fails the traversal test ($R = 0.965$): even genuine structure is not a traversed cycle.

3 Methods

Datasets: GSE134839, GSE150949 (Watermelon), GSE243562 (PDX), GSE131907, and the EGFR-mutant subset of Maynard et al. 2020; standard scanpy QC/normalisation/HVG/PCA (seed 42), subsampled to 1,000–1,500 cells. **Persistent homology:** `ripser` 0.6, Vietoris–Rips to H_1 on top-30 PCs, $\mathbb{F}_2$. **Dispersion null:** draws from $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ of the PCA cloud; ratio = observed/median(null); 30 bootstrap subsamples with 90% CIs; second null by per-PC permutation. **Traversal test:** `dreimac CircularCoords` (DSPVJ), Rayleigh R ; validated on a noisy circle ($R = 0.12$) and a 12%-occupied loop ($R = 0.88$). **Cell-cycle test:** leave-one-cluster-out, Wilcoxon DE with recovered gene symbols, S/G2M regression with the dispersion ratio bootstrapped. **Synthetic benchmark:** four 1,200-cell scenarios, structure in 2-D embedded in 30-D with isotropic noise. **Code:** `scripts/16_*.py`–`scripts/25_*.py` (MIT).

4 Discussion

A cross-system topological signal of drug tolerance—rising $\max\text{-}H_1$ —is largely explained by transcriptional dispersion and, in vitro, by the cell cycle; where genuine non-confound structure remains (in-vivo PDX), it still does not evidence cyclic plasticity, because the population does not traverse it. This is a cautionary, methodological contribution in the established lineage of structure-aware single-cell controls. The biology of persister-state plasticity, established by lineage tracing, is not in question; we show that static persistent-homology summaries do not capture it and instead track nuisance variation. We recommend that topological claims on perturbation scRNA-seq be accompanied, at minimum, by a dispersion-matched null, a traversal test, cell-cycle regression in the dispersion-controlled frame, and bootstrap directional testing.

Outlook. The framework’s ability to isolate (not just reject) structure points to a constructive use: the drug-emergent multiciliated program it surfaced in PDX residual disease is a candidate tolerant-cell state. Establishing it as a discovery would require resolving cell-of-origin (matched-normal CNV or EGFR-driver genotyping), reproduction in further models or patient residual-disease specimens, and functional testing of whether the differentiated state confers tolerance—a prospective, experiment-bearing programme rather than a reanalysis.

Limitations. The cell-cycle attribution is demonstrated on the osimertinib cell line; the PDX residual excess is shown to be non-cell-cycle but its positive identity is not established here. We address $\max\text{-}H_1$; the dispersion null derives from the same filtration and is expected to apply to other persistence summaries (persistence images, Adams et al. 2017; landscapes, Bubenik 2015) and higher-degree homology, though we do not test this. Traversal is inferred from snapshot circular-coordinate concentration plus the clonal-spread proxy, not direct temporal observation.

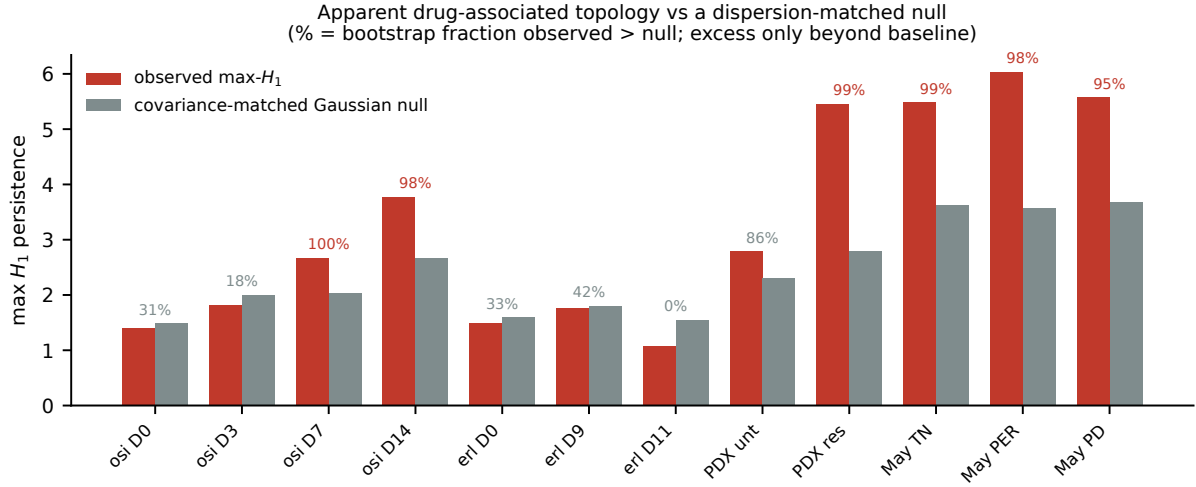


Figure 1: Observed max- H_1 vs a covariance-matched Gaussian (dispersion) null across five systems. Percentages: bootstrap fraction observed > null. Excess appears only under drug.

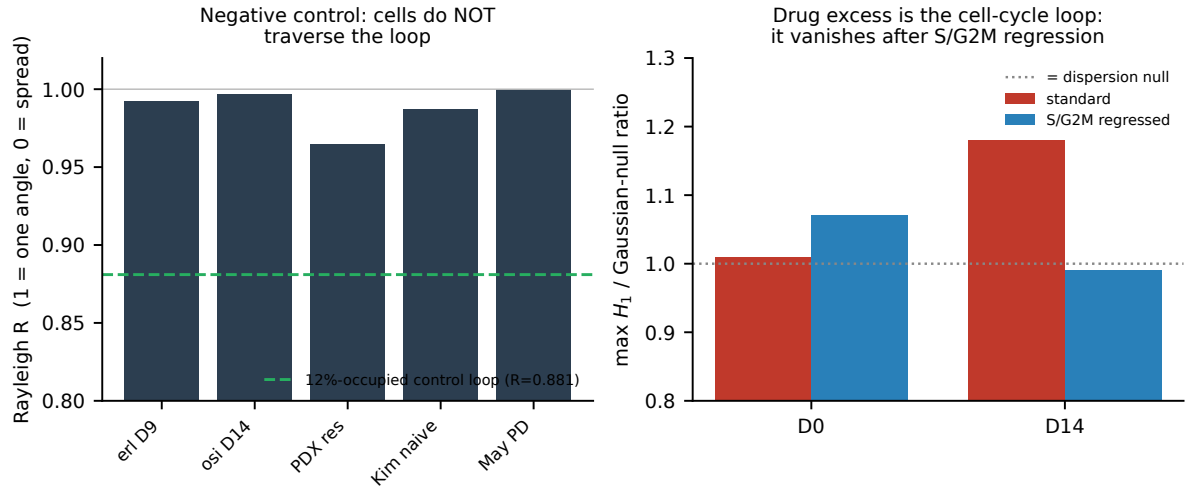


Figure 2: (a) Traversal test: Rayleigh $R \geq 0.965$ in all cohorts vs 0.881 for a 12%-occupied control—cells do not traverse the loop. (b) The in-vitro drug excess vanishes after S/G2M regression.

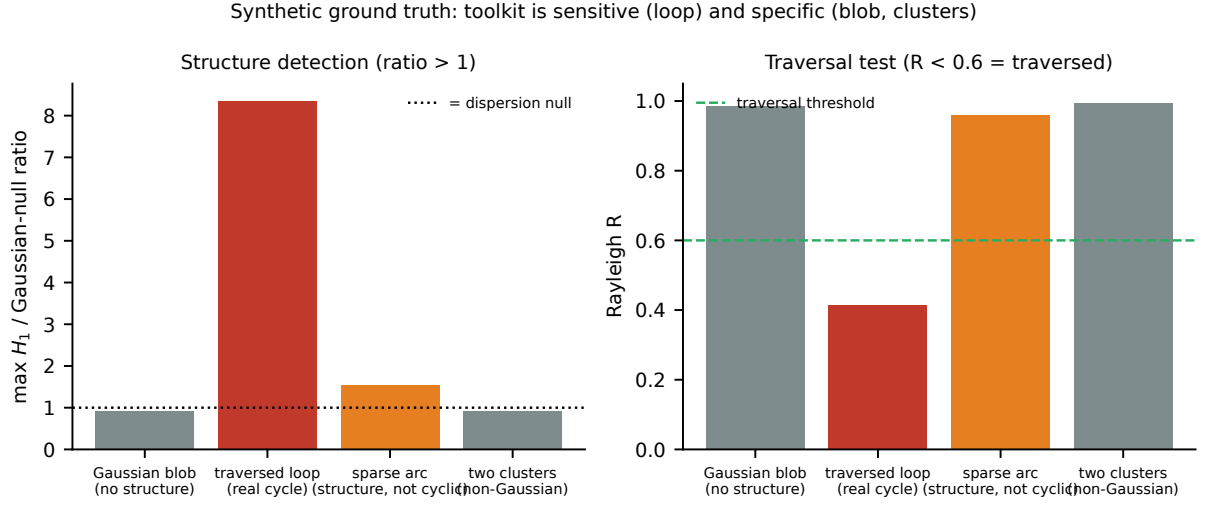


Figure 3: Synthetic ground truth: the framework is sensitive (passes the traversed loop) and specific (rejects the Gaussian blob and the two-cluster non-Gaussian case).

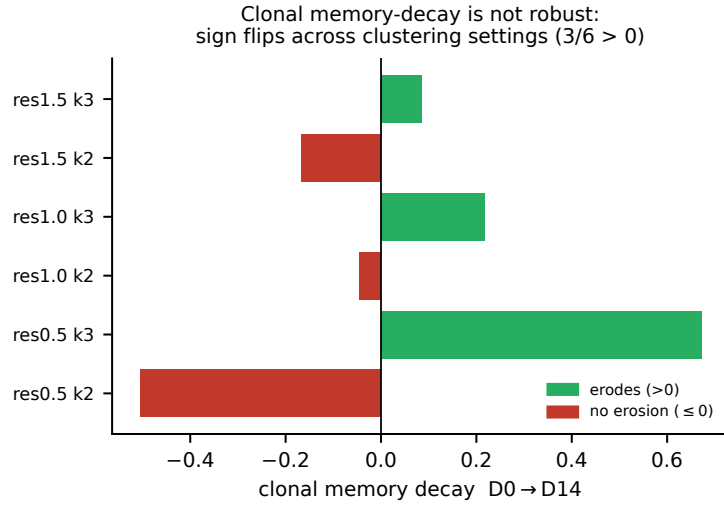


Figure 4: Clonal memory-decay is not robust: the sign of the D0→D14 change flips across clustering settings.